## Marginal and Conditional Distributions

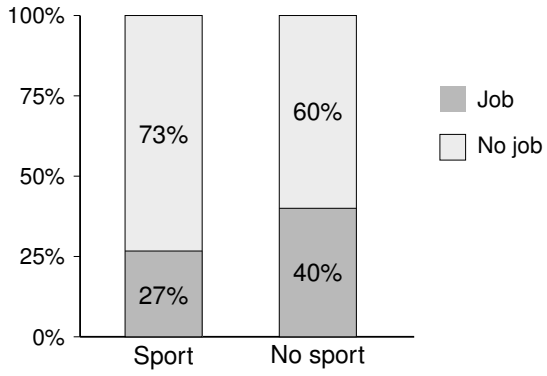
Name _____

Do students who play a sport have jobs as often as students who do not? To answer, compare percents **inside each group**.

| | **Job** | **No job** | **Total** |
|---|---|---|---|
| **Sport** | 8 | 22 | 30 |
| **No sport** | 12 | 18 | 30 |
| **Total** | 20 | 40 | 60 |

| | **What it is** | **Example** |
|---|---|---|
| **Marginal distribution** | Percents from the **totals** (the margins). Divide by the **grand total**. | Job: $\frac{20}{60} \approx 33\%$ No job: $\frac{40}{60} \approx 67\%$ |
| **Conditional distribution** | Percents inside **one group**. Divide by that group's **total**. | Students who play a sport: Job: $\frac{8}{30} \approx 27\%$ No job: $\frac{22}{30} \approx 73\%$ |

**Compare the conditional distributions.**




Sport: $\frac{8}{30} \approx 27\%$ have a job.

No sport: $\frac{12}{30} = 40\%$ have a job.

The percents are **different**. Playing a sport and having a job are **associated**. In this group, students who play a sport are less likely to have a job.

If the percents were about the **same**, there would be **no association**.

**Gotcha: check the denominator.** "Of the students who **play a sport**, what percent have a job?" Divide by the sport total (30), not by 60 and not by the job total (20).

Work with a partner. Compare answers when you finish each problem.

1. Use the table of how 80 students get to school.

| | **Bus** | **Car** | **Walk** | **Total** |
|---|---|---|---|---|
| **9th grade** | 18 | 12 | 10 | 40 |
| **10th grade** | 14 | 20 | 6 | 40 |
| **Total** | 32 | 32 | 16 | 80 |

   Find the marginal distribution of how students get to school.

2. Use the same table.

   (a) Find the conditional distribution of how 9th graders get to school.

   (b) Find the conditional distribution of how 10th graders get to school.

   (c) Is grade associated with how students get to school? Explain.

3. Use the same table.

   (a) What percent of 10th graders come by car?

   (b) What percent of car riders are 10th graders?

   (c) Why are the answers to (a) and (b) different?

4. Is each statement about the school table a **marginal** or a **conditional** percent?

   (a) 40% of all students ride the bus.

   (b) Of the students who walk, 62.5% are 9th graders.

   (c) Half of the students are 10th graders.

   (d) 25% of 9th graders walk.

5. 100 students said if they wear glasses and if they play an instrument.

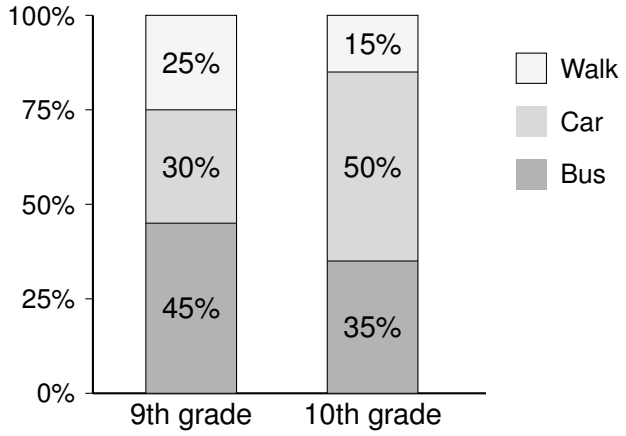
| | **Glasses** | **No glasses** | **Total** |
|---|---|---|---|
| **Instrument** | 12 | 28 | 40 |
| **No instrument** | 18 | 42 | 60 |
| **Total** | 30 | 70 | 100 |

   (a) What percent of students who play an instrument wear glasses?

   (b) What percent of students who do not play an instrument wear glasses?

   (c) Is playing an instrument associated with wearing glasses? Explain.

6. This segmented bar chart shows the conditional distributions from problem 2.




   (a) Which grade has the larger percent of students who walk?

   (b) How does the chart show that grade and transportation are associated?